

Dart Tutorial 1: Introduction and First Program

This chapter introduces Dart in a clear and beginner friendly way. After completing this tutorial, students will understand what Dart is, why it is used in Flutter development, and how to write and run their first Dart program.

1. Introduction to Dart

Dart is a **programming language** developed by Google. It is used to build mobile, web, desktop, and server applications.

Dart is mainly used as the programming language behind **Flutter**, which is used to build cross platform mobile apps.

Dart is an **object oriented**, **type safe**, and **compiled** language.

Dart works very well with Flutter to create fast and smooth user interfaces.

2. Why We Use Dart for Flutter

In real world app development, we need a language that is fast, easy to learn, and works well with UI frameworks.

Dart provides these features:

- Fast compilation using Just In Time (JIT) and Ahead Of Time (AOT) compilation.
- Simple and easy to read syntax, similar to Java and JavaScript.
- Strong support for Object Oriented Programming.
- Built in null safety to avoid runtime errors.
- Hot reload support when used with Flutter.

Because of these features, Dart is the foundation of every Flutter application.

3. Dart vs Other Programming Languages

You might ask — why use Dart when other languages already exist?

Here is the difference:

Feature	Dart	Java	JavaScript
Used For	Flutter Apps	Android Apps	Web Apps
Null Safety	Built In	Limited	Not Built In

Feature	Dart	Java	JavaScript
Compilation	JIT and AOT	JIT/AOT	Interpreted
Learning Curve	Easy	Moderate	Easy
Hot Reload	Yes (with Flutter)	No	No

Dart is specifically optimized for building beautiful and fast user interfaces, which is why Flutter chose it as its core language.

4. Setting Up Dart

Since VS Code and Android Studio setup has already been completed in the Flutter installation tutorials, Dart does not need a separate installation.

Dart comes bundled with the Flutter SDK, so once Flutter is installed, Dart is automatically available.

You can verify Dart is installed by running:

```
dart --version
```

5. Creating Your First Dart File

In any Flutter project, Dart files end with the `.dart` extension.

We create a simple file called `main.dart` to write our first program.

```
void main() {  
  print('Hello Dart');  
}
```

In this example:

- `void main()` is the entry point of every Dart program.
- `print()` is a built-in function used to display output.
- The code inside `{ }` is called the function body.

6. Running Your First Dart Program

There are two common ways to run a Dart program:

Using VS Code Terminal

```
dart run main.dart
```

Using DartPad (Online Editor)

DartPad is a free online tool to write and run Dart code without any setup. It is useful for quick practice.

7. Understanding the Output

When we run the program above, the output will be:

```
Hello Dart
```

This confirms that:

- Dart is correctly installed.
 - The `main()` function executed successfully.
 - The `print()` statement displayed the text on the console.
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8. Basic Structure of a Dart Program

Every Dart program follows a simple structure:

1. Import statements (if required).
2. The `main()` function.
3. Statements and logic inside the function body.

```
void main() {  
  // This is a comment  
  print('Learning Dart Step by Step');  
}
```

9. Why Learning Dart is Important Before Flutter

Although Flutter handles the UI, all the logic behind every app is written in Dart.

- **Widgets** in Flutter are built using Dart classes.
- **State management** depends on Dart variables and functions.

- **App logic** such as calculations and data handling is written in Dart.

This is why learning Dart properly is the first step before becoming a confident Flutter developer.

10. Conclusion of Tutorial 1

In this tutorial, students have learned:

1. What Dart is and why it is important for Flutter.
2. How Dart compares with other programming languages.
3. How Dart is already available through the Flutter SDK.
4. How to create a basic Dart file.
5. How to run a Dart program and understand its output.
6. Why learning Dart is essential before moving into Flutter widgets.

In the next tutorial, we will learn about Variables and Data Types in Dart.

This completes Dart Tutorial 1.